NAME

codata_2010 - Fortran module containing the CODATA 2010 constants

LIBRARY

codata (**-libcodata**, **-lcodata**)

SYNOPSIS

```
use codata

include "codata.h"

import pycodata
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DESCRIPTION

List of available constants:

- LATTICE_SPACING_OF_SILICON_2010
- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2010
- ALPHA_PARTICLE_MASS_2010
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2010
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- ALPHA_PARTICLE_MASS_IN_U_2010
- ALPHA_PARTICLE_MOLAR_MASS_2010
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2010
- ANGSTROM_STAR_2010
- ATOMIC_MASS_CONSTANT_2010
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2010
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2010
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2010
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2010
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2010
- ATOMIC_UNIT_OF_ACTION_2010
- ATOMIC_UNIT_OF_CHARGE_2010
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2010
- ATOMIC_UNIT_OF_CURRENT_2010
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2010
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2010

- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2010
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2010
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2010
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2010
- ATOMIC_UNIT_OF_ENERGY_2010
- ATOMIC_UNIT_OF_FORCE_2010
- ATOMIC_UNIT_OF_LENGTH_2010
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2010
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2010
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2010
- ATOMIC_UNIT_OF_MASS_2010
- ATOMIC_UNIT_OF_MOMUM_2010
- ATOMIC_UNIT_OF_PERMITTIVITY_2010
- ATOMIC_UNIT_OF_TIME_2010
- ATOMIC_UNIT_OF_VELOCITY_2010
- AVOGADRO_CONSTANT_2010
- BOHR_MAGNETON_2010
- BOHR_MAGNETON_IN_EV_T_2010
- BOHR_MAGNETON_IN_HZ_T_2010
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010
- BOHR_MAGNETON_IN_K_T_2010
- BOHR_RADIUS_2010
- BOLTZMANN_CONSTANT_2010
- BOLTZMANN_CONSTANT_IN_EV_K_2010
- BOLTZMANN_CONSTANT_IN_HZ_K_2010
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2010
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2010
- CLASSICAL_ELECTRON_RADIUS_2010
- COMPTON_WAVELENGTH_2010
- COMPTON_WAVELENGTH_OVER_2_PI_2010
- CONDUCTANCE_QUANTUM_2010
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2010
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2010
- CU_X_UNIT_2010
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2010
- DEUTERON_ELECTRON_MASS_RATIO_2010
- DEUTERON_G_FACTOR_2010
- DEUTERON_MAG_MOM_2010

- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- DEUTERON_MASS_2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- DEUTERON_MASS_IN_U_2010
- DEUTERON_MOLAR_MASS_2010
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MASS_RATIO_2010
- DEUTERON_RMS_CHARGE_RADIUS_2010
- ELECTRIC_CONSTANT_2010
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2010
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2010
- ELECTRON_DEUTERON_MASS_RATIO_2010
- ELECTRON_G_FACTOR_2010
- ELECTRON_GYROMAG_RATIO_2010
- ELECTRON_GYROMAG_RATIO_OVER_2_PI_2010
- ELECTRON_HELION_MASS_RATIO_2010
- ELECTRON_MAG_MOM_2010
- ELECTRON_MAG_MOM_ANOMALY_2010
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- ELECTRON_MASS_2010
- ELECTRON_MASS_ENERGY_EQUIVALENT_2010
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- ELECTRON_MASS_IN_U_2010
- ELECTRON_MOLAR_MASS_2010
- ELECTRON_MUON_MAG_MOM_RATIO_2010
- ELECTRON_MUON_MASS_RATIO_2010
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2010
- ELECTRON_NEUTRON_MASS_RATIO_2010
- ELECTRON_PROTON_MAG_MOM_RATIO_2010
- ELECTRON_PROTON_MASS_RATIO_2010
- ELECTRON_TAU_MASS_RATIO_2010
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2010
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2010
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010

- ELECTRON_TRITON_MASS_RATIO_2010
- ELECTRON_VOLT_2010
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2010
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2010
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2010
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2010
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- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2010
- ELEMENTARY_CHARGE_2010
- ELEMENTARY_CHARGE_OVER_H_2010
- FARADAY_CONSTANT_2010
- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2010
- FERMI_COUPLING_CONSTANT_2010
- FINE_STRUCTURE_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2010
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2010
- HARTREE_ENERGY_2010
- HARTREE_ENERGY_IN_EV_2010
- HARTREE_HERTZ_RELATIONSHIP_2010
- HARTREE_INVERSE_METER_RELATIONSHIP_2010
- HARTREE_JOULE_RELATIONSHIP_2010
- HARTREE_KELVIN_RELATIONSHIP_2010
- HARTREE_KILOGRAM_RELATIONSHIP_2010
- HELION_ELECTRON_MASS_RATIO_2010
- HELION_G_FACTOR_2010
- HELION_MAG_MOM_2010
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- HELION_MASS_2010
- HELION_MASS_ENERGY_EQUIVALENT_2010
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- HELION_MASS_IN_U_2010
- HELION_MOLAR_MASS_2010
- HELION_PROTON_MASS_RATIO_2010
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2010

- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2010
- HERTZ_HARTREE_RELATIONSHIP_2010
- HERTZ_INVERSE_METER_RELATIONSHIP_2010
- HERTZ_JOULE_RELATIONSHIP_2010
- HERTZ_KELVIN_RELATIONSHIP_2010
- HERTZ_KILOGRAM_RELATIONSHIP_2010
- INVERSE_FINE_STRUCTURE_CONSTANT_2010
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2010
- INVERSE_METER_HARTREE_RELATIONSHIP_2010
- INVERSE_METER_HERTZ_RELATIONSHIP_2010
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- INVERSE_METER_KILOGRAM_RELATIONSHIP_2010
- INVERSE_OF_CONDUCTANCE_QUANTUM_2010
- JOSEPHSON_CONSTANT_2010
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2010
- JOULE_HARTREE_RELATIONSHIP_2010
- JOULE_HERTZ_RELATIONSHIP_2010
- JOULE_INVERSE_METER_RELATIONSHIP_2010
- JOULE_KELVIN_RELATIONSHIP_2010
- JOULE_KILOGRAM_RELATIONSHIP_2010
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2010
- KELVIN_HARTREE_RELATIONSHIP_2010
- KELVIN_HERTZ_RELATIONSHIP_2010
- KELVIN_INVERSE_METER_RELATIONSHIP_2010
- KELVIN_JOULE_RELATIONSHIP_2010
- KELVIN_KILOGRAM_RELATIONSHIP_2010
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2010
- KILOGRAM_HARTREE_RELATIONSHIP_2010
- KILOGRAM_HERTZ_RELATIONSHIP_2010
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2010
- KILOGRAM_JOULE_RELATIONSHIP_2010
- KILOGRAM_KELVIN_RELATIONSHIP_2010
- LATTICE_PARAMETER_OF_SILICON_2010

- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2010
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2010
- MAG_CONSTANT_2010
- MAG_FLUX_QUANTUM_2010
- MOLAR_GAS_CONSTANT_2010
- MOLAR_MASS_CONSTANT_2010
- MOLAR_MASS_OF_CARBON_12_2010
- MOLAR_PLANCK_CONSTANT_2010
- MOLAR_PLANCK_CONSTANT_TIMES_C_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2010
- MOLAR_VOLUME_OF_SILICON_2010
- MO_X_UNIT_2010
- MUON_COMPTON_WAVELENGTH_2010
- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- MUON_ELECTRON_MASS_RATIO_2010
- MUON_G_FACTOR_2010
- MUON_MAG_MOM_2010
- MUON_MAG_MOM_ANOMALY_2010
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- MUON_MASS_2010
- MUON_MASS_ENERGY_EQUIVALENT_2010
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
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- MUON_NEUTRON_MASS_RATIO_2010
- MUON_PROTON_MAG_MOM_RATIO_2010
- MUON_PROTON_MASS_RATIO_2010
- MUON_TAU_MASS_RATIO_2010
- NATURAL_UNIT_OF_ACTION_2010
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2010
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- NATURAL_UNIT_OF_ENERGY_IN_MEV_2010
- NATURAL_UNIT_OF_LENGTH_2010
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- NATURAL_UNIT_OF_MOMUM_2010
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C_2010

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- NATURAL_UNIT_OF_VELOCITY_2010
- NEUTRON_COMPTON_WAVELENGTH_2010
- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2010
- NEUTRON_ELECTRON_MASS_RATIO_2010
- NEUTRON_G_FACTOR_2010
- NEUTRON_GYROMAG_RATIO_2010
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2010
- NEUTRON_MAG_MOM_2010
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- NEUTRON_MASS_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_MASS_IN_U_2010
- NEUTRON_MOLAR_MASS_2010
- NEUTRON_MUON_MASS_RATIO_2010
- NEUTRON_PROTON_MAG_MOM_RATIO_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2010
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- NEUTRON_PROTON_MASS_RATIO_2010
- NEUTRON_TAU_MASS_RATIO_2010
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- NEWTONIAN_CONSTANT_OF_GRAVITATION_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2010
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- NUCLEAR_MAGNETON_IN_EV_T_2010
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- PLANCK_CONSTANT_2010
- PLANCK_CONSTANT_IN_EV_S_2010
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- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2010
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- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2010
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- PROTON_MASS_2010
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- PROTON_MASS_IN_U_2010
- PROTON_MOLAR_MASS_2010
- PROTON_MUON_MASS_RATIO_2010
- PROTON_NEUTRON_MAG_MOM_RATIO_2010
- PROTON_NEUTRON_MASS_RATIO_2010
- PROTON_RMS_CHARGE_RADIUS_2010
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- QUANTUM_OF_CIRCULATION_2010
- QUANTUM_OF_CIRCULATION_TIMES_2_2010
- RYDBERG_CONSTANT_2010
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2010
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2010
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- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2010
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- SECOND_RADIATION_CONSTANT_2010
- SHIELDED_HELION_GYROMAG_RATIO_2010
- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2010
- SHIELDED_HELION_MAG_MOM_2010

- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2010
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
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- SHIELDED_PROTON_MAG_MOM_2010
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
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- TRITON_PROTON_MASS_RATIO_2010
- UNIFIED_ATOMIC_MASS_UNIT_2010

- VON_KLITZING_CONSTANT_2010
- WEAK_MIXING_ANGLE_2010
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2010
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2010

SEE ALSO

codata(3), **codata_2010**(3), **codata_2014**(3), **codata_2018**(3), **codata_2022**(3)